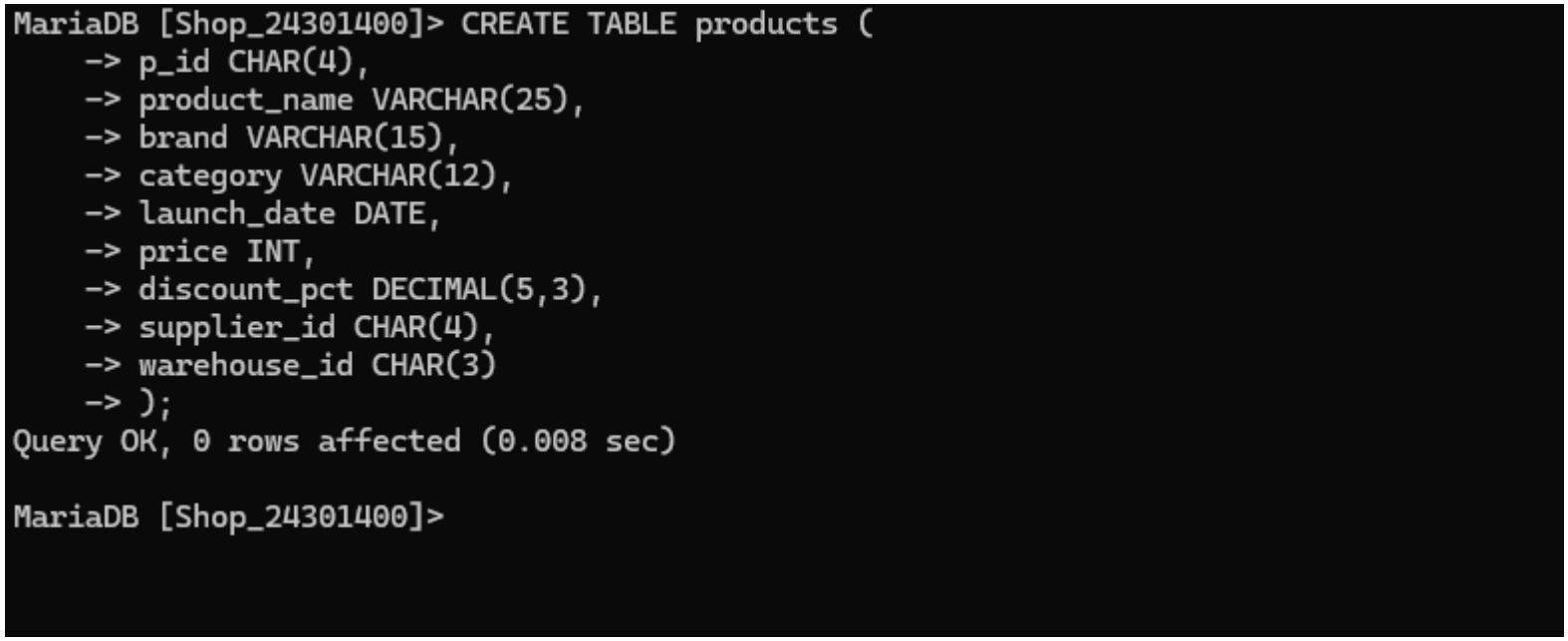
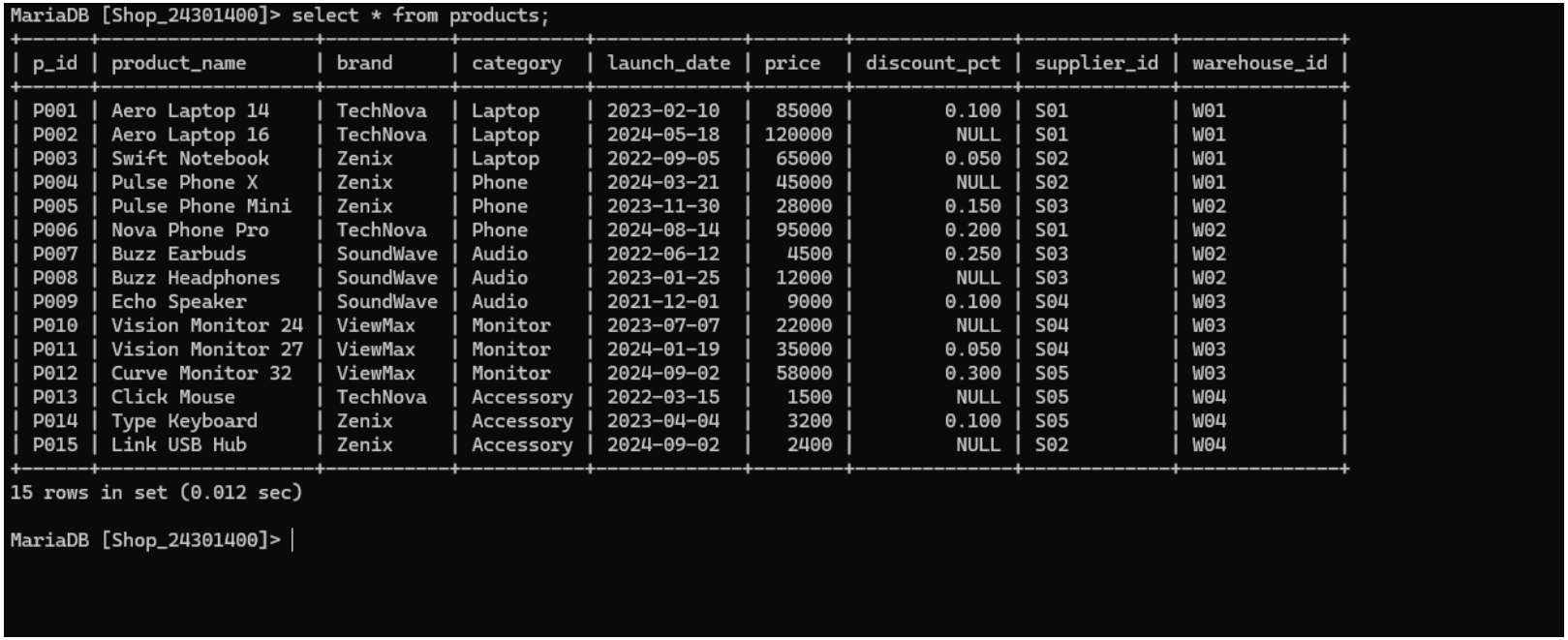
CSE370 : Database Systems

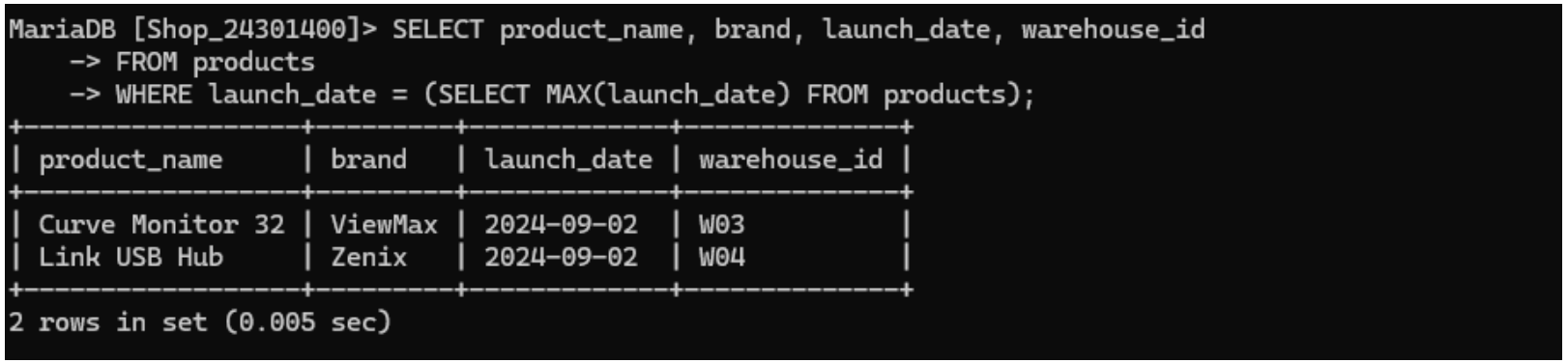
Assignment 01/02/03 | Fall 2026

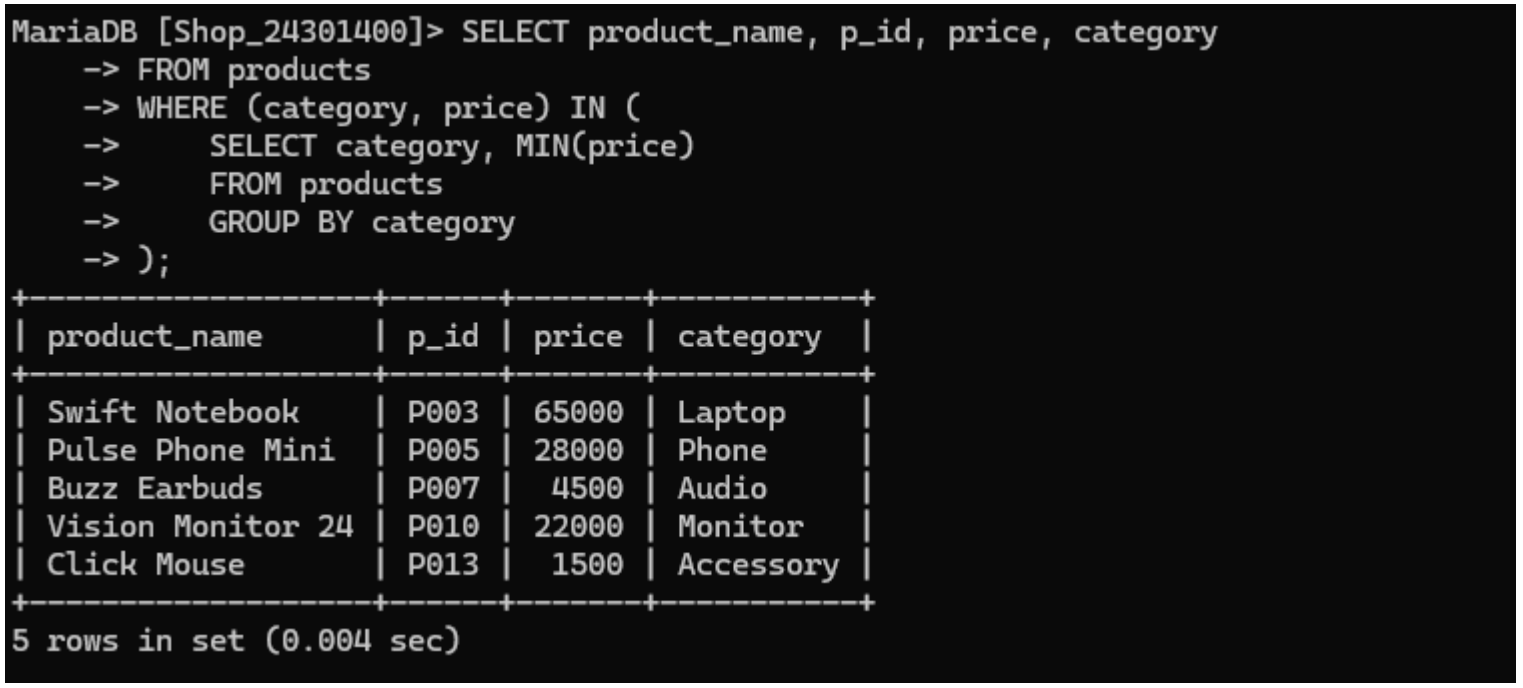
ID : <24301400> | Name : Md. Mehtabul Islam

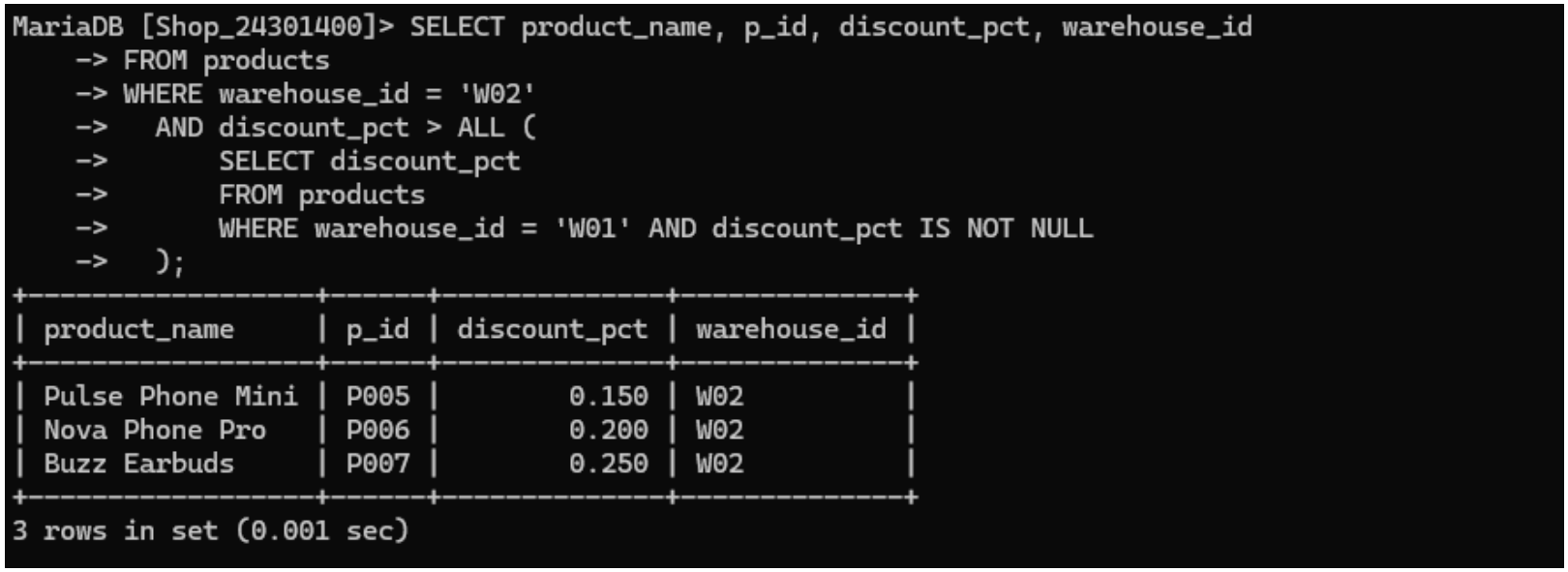
No 1 Query (as Plain Text)	CREATE TABLE products (-> p_id CHAR(4), -> product_name VARCHAR(25), -> brand VARCHAR(15), -> category VARCHAR(12), -> launch_date DATE, -> price INT, -> discount_pct DECIMAL(5,3), -> supplier_id CHAR(4), -> warehouse_id CHAR(3));

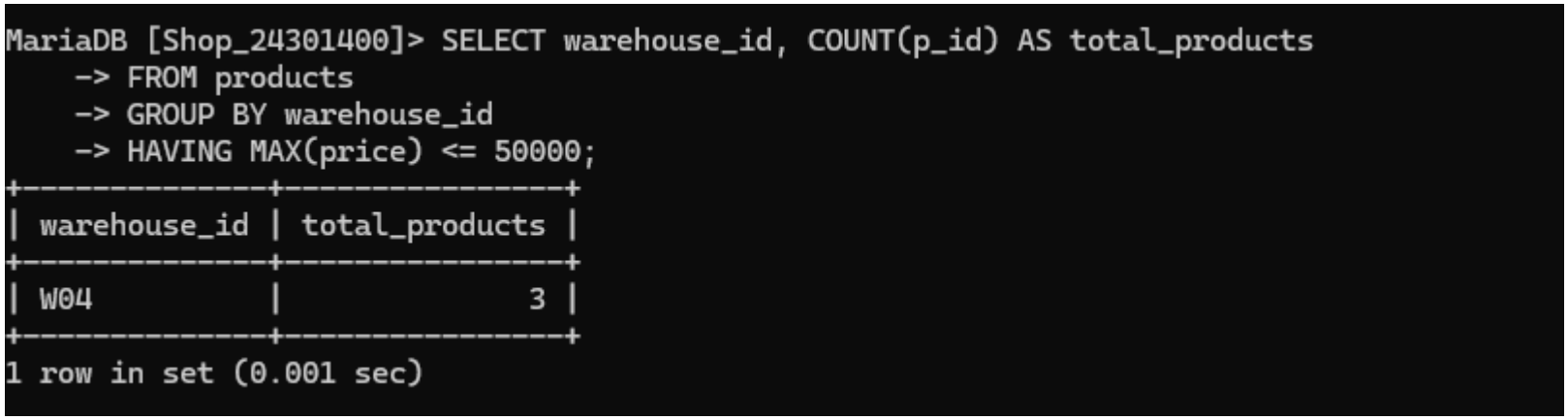
No 1 SS **(of Query & Output in Shell)**	
No 2 Query **(as Plain Text)**	INSERT INTO products (p_id, product_name, brand, category, launch_date, price, discount_pct, supplier_id, warehouse_id) VALUES -> ('P001', 'Aero Laptop 14', 'TechNova', 'Laptop', '2023-02-10', 85000, 0.100, 'S01', 'W01'), -> ('P002', 'Aero Laptop 16', 'TechNova', 'Laptop', '2024-05-18', 120000, NULL, 'S01', 'W01'), -> ('P003', 'Swift Notebook', 'Zenix', 'Laptop', '2022-09-05', 65000, 0.050, 'S02', 'W01'), -> ('P004', 'Pulse Phone X', 'Zenix', 'Phone', '2024-03-21', 45000, NULL, 'S02', 'W01'), -> ('P005', 'Pulse Phone Mini', 'Zenix', 'Phone', '2023-11-30', 28000, 0.150, 'S03', 'W02'), -> ('P006', 'Nova Phone Pro', 'TechNova', 'Phone', '2024-08-14', 95000, 0.200, 'S01', 'W02'), -> ('P007', 'Buzz Earbuds', 'SoundWave', 'Audio', '2022-06-12', 4500, 0.250, 'S03', 'W02'), -> ('P008', 'Buzz Headphones', 'SoundWave', 'Audio', '2023-01-25', 12000, NULL, 'S03', 'W02'), -> ('P009', 'Echo Speaker', 'SoundWave', 'Audio', '2021-12-01', 9000, 0.100, 'S04', 'W03'), -> ('P010', 'Vision Monitor 24', 'ViewMax', 'Monitor', '2023-07-07', 22000, NULL, 'S04', 'W03'),

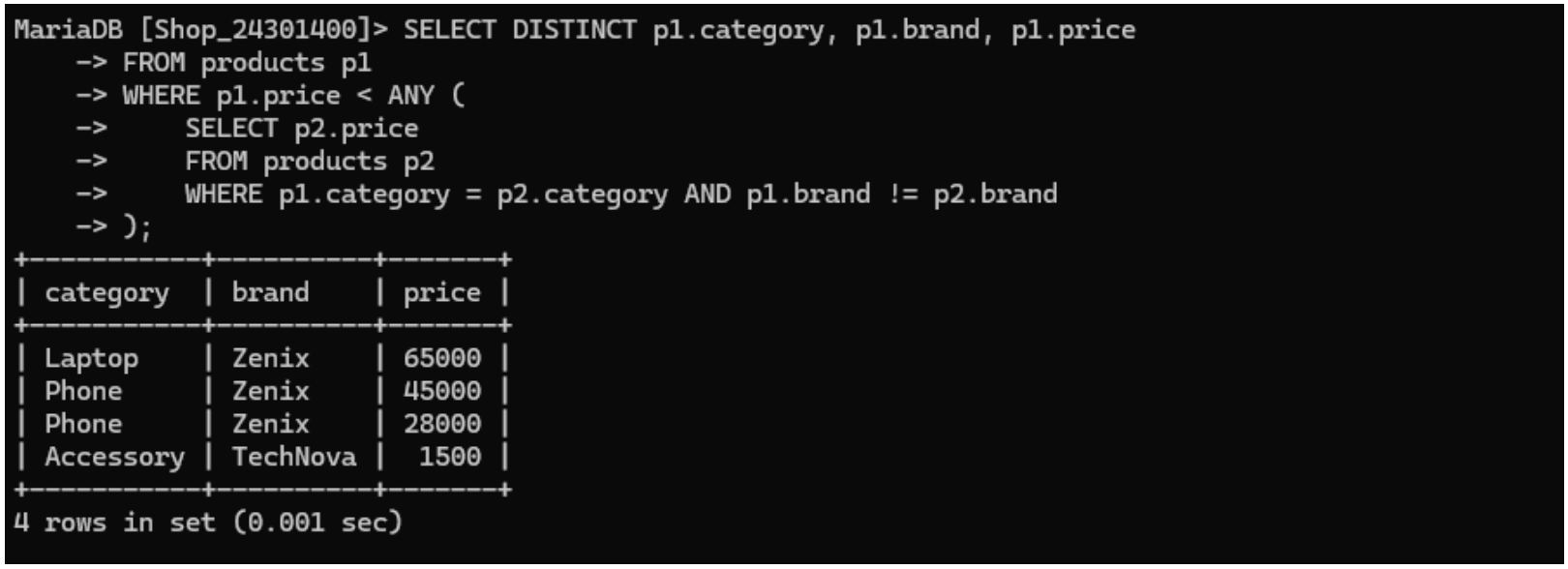
	-> ('P011', 'Vision Monitor 27', 'ViewMax', 'Monitor', '2024-01-19', 35000, 0.050, 'S04', 'W03'), -> ('P012', 'Curve Monitor 32', 'ViewMax', 'Monitor', '2024-09-02', 58000, 0.300, 'S05', 'W03'), -> ('P013', 'Click Mouse', 'TechNova', 'Accessory', '2022-03-15', 1500, NULL, 'S05', 'W04'), -> ('P014', 'Type Keyboard', 'Zenix', 'Accessory', '2023-04-04', 3200, 0.100, 'S05', 'W04'), -> ('P015', 'Link USB Hub', 'Zenix', 'Accessory', '2024-09-02', 2400, NULL, 'S02', 'W04');
No 2 SS (of Query & Output in Shell)	
No 3 Query (as Plain Text)	SELECT product_name, brand, launch_date, warehouse_id -> FROM products -> WHERE launch_date = (SELECT MAX(launch_date) FROM products);

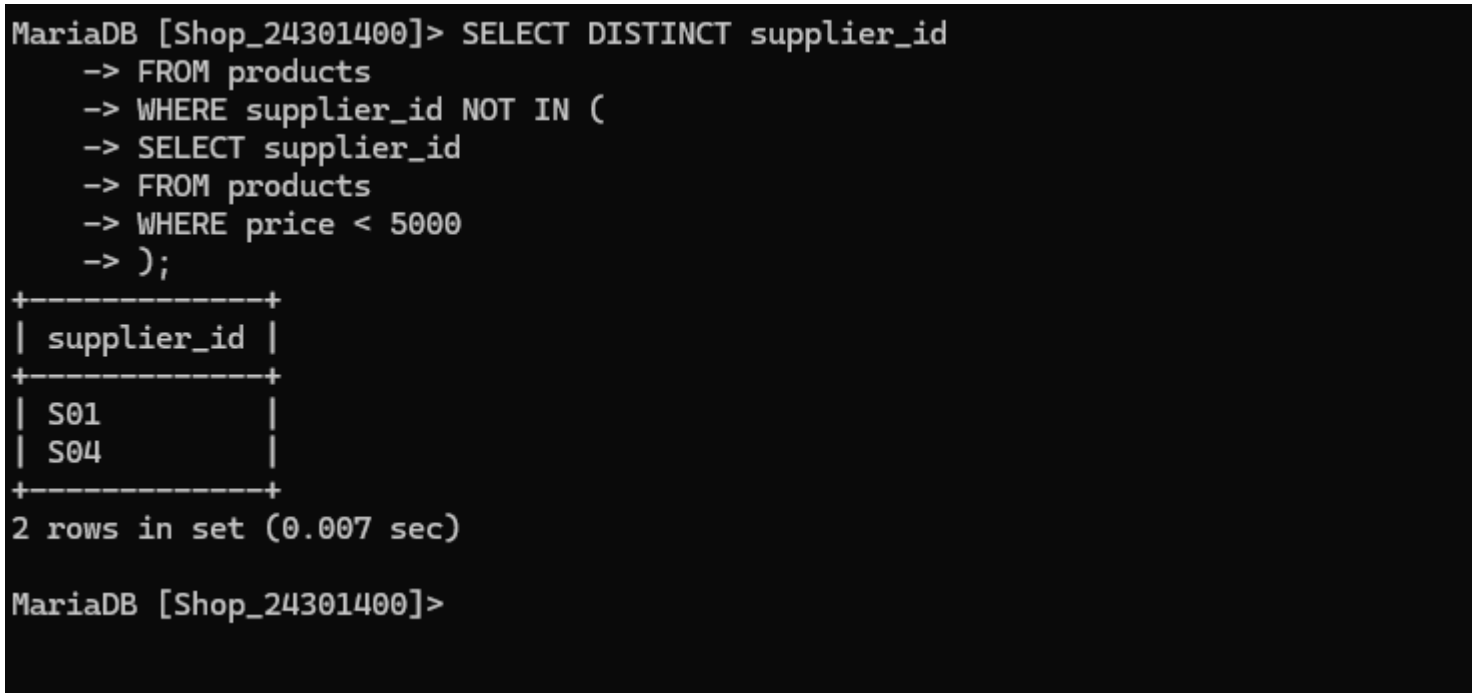
No 3 SS **(of Query & Output in Shell)**	
No 4 Query **(as Plain Text)**	SELECT product_name, p_id, price, category -> FROM products -> WHERE (category, price) IN (-> SELECT category, MIN(price) -> FROM products -> GROUP BY category ->);

No 4 SS (of Query & Output in Shell)	
No 5 Query (as Plain Text)	SELECT product_name, p_id, discount_pct, warehouse_id -> FROM products -> WHERE warehouse_id = 'W02' -> AND discount_pct > ALL (-> SELECT discount_pct -> FROM products -> WHERE warehouse_id = 'W01' AND discount_pct IS NOT NULL ->);

No 5 SS **(of Query & Output in Shell)**	
No 6 Query **(as Plain Text)**	SELECT warehouse_id, COUNT(p_id) AS total_products -> FROM products -> GROUP BY warehouse_id -> HAVING MAX(price) <= 50000;

No 6 SS **(of Query & Output in Shell)**	
No 7 Query **(as Plain Text)**	SELECT DISTINCT p1.category, p1.brand, p1.price -> FROM products p1 -> WHERE p1.price < ANY (-> SELECT p2.price -> FROM products p2 -> WHERE p1.category = p2.category AND p1.brand != p2.brand ->);

No 7 SS **(of Query & Output in Shell)**	
No 8 Query **(as Plain Text)**	SELECT DISTINCT supplier_id -> FROM products -> WHERE supplier_id NOT IN (-> SELECT supplier_id -> FROM products -> WHERE price < 5000 ->);

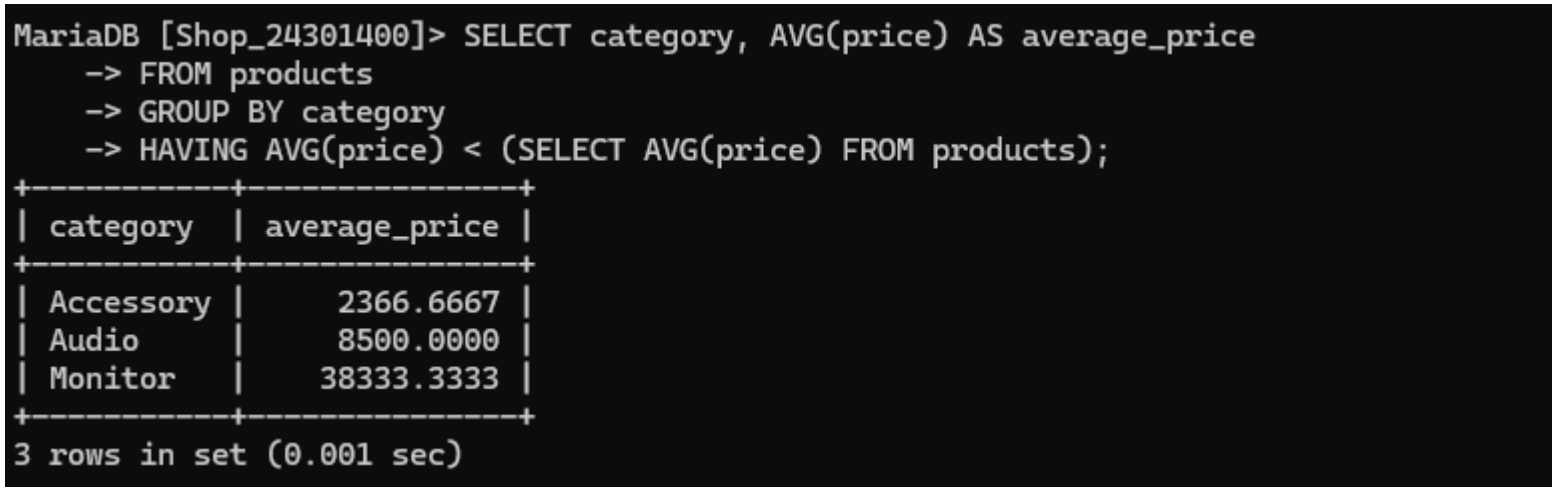
No 8 SS (of Query & Output in Shell)	
No 9 Query (as Plain Text)	SELECT product_name, p_id, price, warehouse_id, discount_pct -> FROM products -> WHERE (supplier_id, discount_pct) IN (-> SELECT supplier_id, MAX(discount_pct) -> FROM products -> GROUP BY supplier_id ->);

No 9 SS
(of Query & Output in Shell)

```
MariaDB [Shop_24301400]> SELECT product_name, p_id, price, warehouse_id, discount_pct
    -> FROM products
    -> WHERE (supplier_id, discount_pct) IN (
    ->     SELECT supplier_id, MAX(discount_pct)
    ->     FROM products
    ->     GROUP BY supplier_id
    -> );
+-----------------+------+-------+--------------+--------------+
| product_name    | p_id | price | warehouse_id | discount_pct |
+-----------------+------+-------+--------------+--------------+
| Swift Notebook  | P003 | 65000 | W01          |        0.050 |
| Nova Phone Pro  | P006 | 95000 | W02          |        0.200 |
| Buzz Earbuds    | P007 |  4500 | W02          |        0.250 |
| Echo Speaker    | P009 |  9000 | W03          |        0.100 |
| Curve Monitor 32| P012 | 58000 | W03          |        0.300 |
+-----------------+------+-------+--------------+--------------+
5 rows in set (0.001 sec)
```

No 10 Query
(as Plain Text)

```
SELECT category, AVG(price) AS average_price
   -> FROM products
   -> GROUP BY category
   -> HAVING AVG(price) < (SELECT AVG(price) FROM products);
```

No 10 SS (of Query & Output in Shell)	

```
MariaDB [Shop_24301400]> SELECT category, AVG(price) AS average_price
    -> FROM products
    -> GROUP BY category
    -> HAVING AVG(price) < (SELECT AVG(price) FROM products);
+-----------+---------------+
| category  | average_price |
+-----------+---------------+
| Accessory |     2366.6667 |
| Audio     |     8500.0000 |
| Monitor   |    38333.3333 |
+-----------+---------------+
3 rows in set (0.001 sec)
```